

CHURCH FENTON, United Kingdom

WMO: 033550

Lat: 53.834N

Lon: 1.196W

Elev: 9

StdP: 101.22

Time Zone: 0.00 (EUW)

Period: 01-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-4.6	-2.9	-6.7	2.1	-1.3	-5.3	2.4	-0.7	14.7	9.6	13.3	9.9	2.0	330	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	25.4	18.3	23.5	17.4	21.9	16.4	19.2	23.5	18.1	22.2	17.2	20.7	4.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	12.6	20.9	16.6	11.8	19.7	15.7	11.1	18.8	54.8	23.6	51.0	22.1	48.3	20.7	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
12.3	10.6	9.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-8.2	28.7	3.2	1.6	-10.5	29.8	-12.3	30.8	-14.1	31.7	-16.4	32.9		
			-8.5	22.2	3.2	3.1	-10.8	24.4	-12.6	26.3	-14.4	28.0	-16.7	30.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	4.6	4.5	6.2	8.7	11.5	14.6	16.5	16.3	14.1	10.6	6.9	3.9
	DBStd	5.37	3.43	3.47	2.98	2.81	2.75	2.47	2.39	2.15	2.41	2.89	3.25	3.79
	HDD10.0	838	168	156	122	57	16	1	0	0	1	27	101	190
	HDD18.3	3104	425	387	377	289	213	116	69	72	129	239	343	446
	CDD10.0	798	1	2	4	18	61	138	201	194	123	46	8	2
	CDD18.3	23	0	0	0	0	0	3	12	7	1	0	0	0
	CDH23.3	177	0	0	0	0	5	28	94	36	12	2	0	0
	CDH26.7	18	0	0	0	0	0	3	8	6	1	1	0	0
Wind	WSAvg	4.3	5.1	4.6	4.6	4.3	4.4	3.8	3.7	3.9	4.0	4.2	4.5	4.4
Precipitation	PrecAvg	586	49	37	46	43	47	51	46	60	52	48	52	55
	PrecMax	732	89	116	104	95	121	145	103	130	150	116	108	161
	PrecMin	397	11	1	7	7	10	6	6	9	8	11	16	22
	PrecStd	90	22	25	24	24	28	32	26	31	36	25	24	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	14.0	16.2	20.5	23.8	26.7	27.6	27.3	24.9	19.6	15.5	13.7
		MCWB	10.9	10.8	10.9	13.7	17.6	19.6	18.8	20.9	16.5	15.6	13.3	11.2
	2%	DB	12.1	12.3	14.3	17.7	20.8	23.5	25.9	23.8	22.1	17.4	13.9	12.2
		MCWB	9.9	10.1	10.0	11.8	14.9	16.9	18.7	18.1	16.4	14.4	11.7	10.2
	5%	DB	10.9	11.0	12.8	15.9	18.7	21.7	24.0	22.1	20.4	16.2	13.0	11.0
		MCWB	8.9	8.7	9.5	11.1	13.5	16.1	17.7	16.9	15.4	13.6	11.0	9.3
	10%	DB	9.8	9.8	11.4	14.4	16.8	19.9	21.7	20.8	18.8	15.1	11.8	9.5
		MCWB	8.0	7.6	8.5	10.2	12.3	15.1	16.3	16.1	14.5	12.6	10.0	8.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	11.7	12.0	14.6	17.4	20.4	20.7	22.0	18.5	16.6	13.8	11.5
		MCDB	13.0	13.3	14.7	18.4	23.0	25.0	25.5	25.0	22.5	19.2	15.3	13.0
	2%	WB	10.2	10.2	10.8	13.1	15.5	18.1	19.1	19.1	17.1	15.2	12.3	10.5
		MCDB	11.8	12.1	13.6	16.5	20.0	21.8	24.1	22.4	20.5	17.0	13.6	11.9
	5%	WB	9.2	9.1	9.8	11.6	14.1	16.8	18.0	17.9	16.1	14.1	11.2	9.6
		MCDB	10.6	10.8	12.4	15.0	17.3	20.4	23.0	21.0	19.0	15.7	12.5	10.8
	10%	WB	8.1	8.0	8.7	10.6	13.0	15.6	17.1	17.0	15.3	13.1	10.2	8.3
		MCDB	9.6	9.5	11.3	13.8	16.1	19.2	20.9	19.6	18.1	14.7	11.6	9.5
Mean Daily Temperature Range	5% DB	MDBR	5.2	6.1	7.7	9.1	8.9	8.8	8.8	8.3	8.4	6.9	6.1	5.5
		MCDBR	6.4	7.0	10.0	11.7	12.1	12.3	12.7	10.5	11.0	7.8	6.5	6.1
		MCWBR	5.5	5.5	6.3	6.9	6.8	6.8	6.2	5.6	6.1	5.3	5.4	5.4
	5% WB	MCDBR	6.5	6.6	9.2	11.0	11.2	11.1	12.0	9.0	9.5	7.2	6.4	6.0
		MCWBR	5.7	5.4	6.3	7.3	6.7	7.0	6.3	5.6	6.0	5.5	5.6	5.4
Clear-Sky Solar Irradiance	taub	0.321	0.337	0.368	0.396	0.395	0.400	0.401	0.388	0.368	0.349	0.323	0.309	
	taud	2.342	2.385	2.301	2.237	2.278	2.290	2.293	2.349	2.413	2.442	2.398	2.318	
	Ebn at Noon	636	757	805	830	852	849	840	831	802	734	633	569	
	Edh at Noon	58	76	101	122	124	124	121	108	90	71	54	49	
All-Sky Solar Radiation	RadAvg	0.64	1.30	2.38	3.72	4.70	4.83	4.69	3.86	2.83	1.60	0.79	0.51	
	RadStd	0.07	0.15	0.29	0.43	0.29	0.44	0.46	0.25	0.26	0.14	0.08	0.05	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page